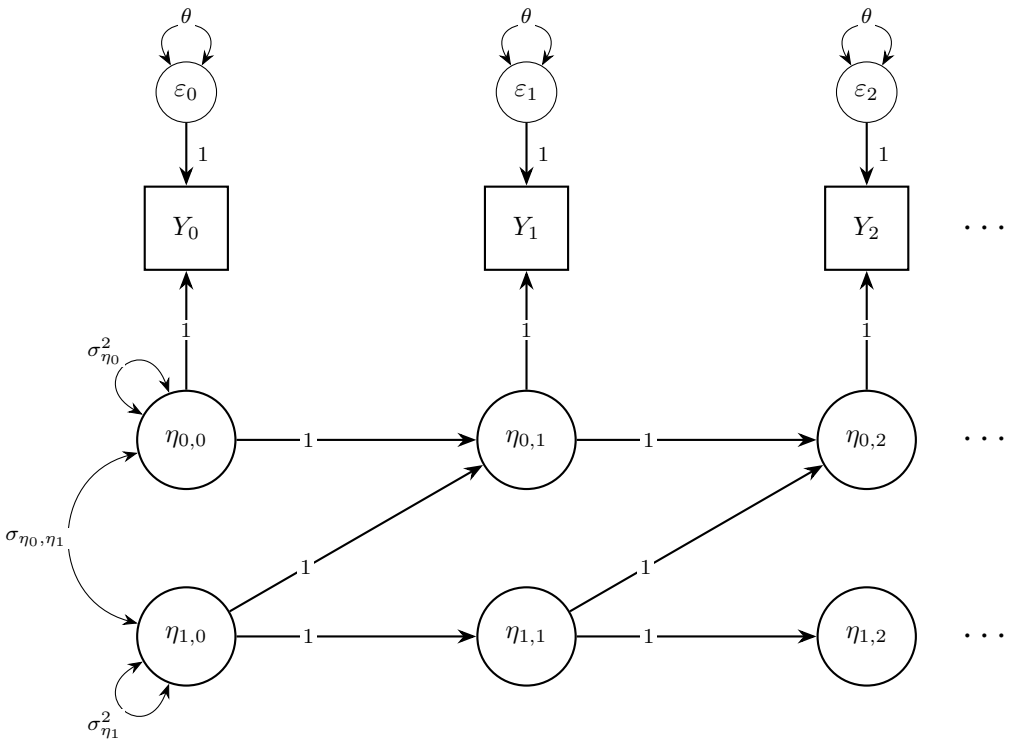



initial condition

$$\boldsymbol{\eta}_t = (\eta_{0,t}, \eta_{1,t})' \text{ (level, slope)}, \quad \mathbf{y}_t = Y_t \text{; no process noise}$$

$$\boldsymbol{\beta} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \qquad \boldsymbol{\Lambda} = \begin{pmatrix} 1 & 0 \end{pmatrix} \qquad \boldsymbol{\Theta} = \theta$$

$$\boldsymbol{\eta}_0 \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0), \quad \boldsymbol{\mu}_0 = \begin{pmatrix} \mu_{\eta_0} \\ \mu_{\eta_1} \end{pmatrix}, \quad \boldsymbol{\Sigma}_0 = \begin{pmatrix} \sigma^2_{\eta_0} & \sigma_{\eta_0,\eta_1} \\ \sigma_{\eta_0,\eta_1} & \sigma^2_{\eta_1} \end{pmatrix}$$

$$\eta_{0,0} = \eta_0 \text{ (intercept)}, \quad \eta_{1,t} = \eta_1 \text{ for all } t \text{ (slope)}, \quad \eta_{0,t} = \eta_0 + t\eta_1$$